

1. Sketch the following signals:

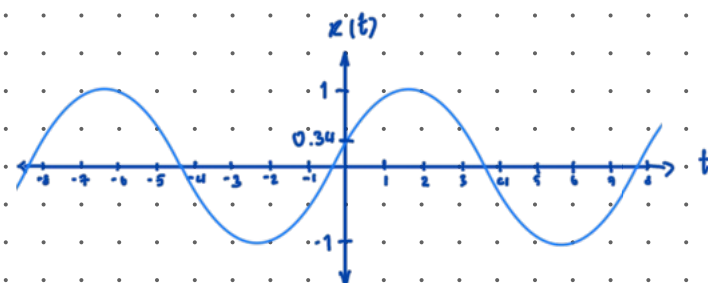
นักศึกษา: วิริยะ เกียรติพนาศ
6730054721

1.1 $x(t) = \sin\left(\frac{\pi}{4}t + \frac{\pi}{4}\right)$

Period is $T = \frac{2\pi}{|\omega_0|} = \frac{2\pi}{\pi/4} = 8$

When $t = 0$; $x(t) = \sin\left(\frac{\pi}{4}\right) \approx 0.34$

Thus, we obtain a graph something like on the right.

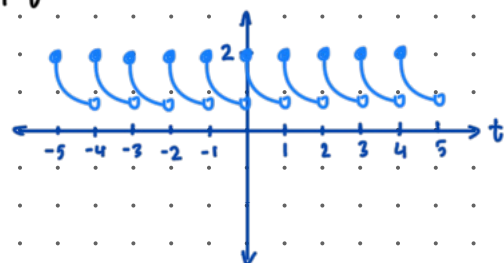


1.2 $x(t) = 2e^{-t}$, $0 \leq t < 1$ and $x(t+1) = x(t)$ for all t

when $t = 0$, $x(t) = 2$

$t = 1^-$, $x(t) = 2e^{-1} \approx 0.74$

Since $x(t+1) = x(t)$, we obtain a graph on the right.



1.5 $x(t) = r(t) - r(t-1) - u(t-2)$

$$r(t) = \begin{cases} 0; & t < 0 \\ t; & t \geq 0 \end{cases}$$

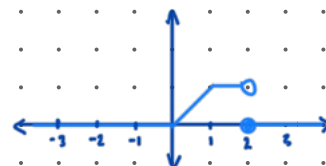
$$u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases}$$

When $t < 0$; $x(t) = 0 - 0 - 0 = 0$

$0 \leq t < 1$; $x(t) = t - 0 - 0 = t$

$1 \leq t < 2$; $x(t) = t - (t-1) - 0 = 1$

$t \geq 2$; $x(t) = t - (t-1) - 1 = 0$

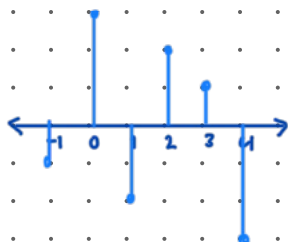


4. For the discrete time signal $x[n]$ shown in figure, can be written as:

n	-2	-1	0	1	2	3
$x[n]$	-3	1	2	-2	3	-1

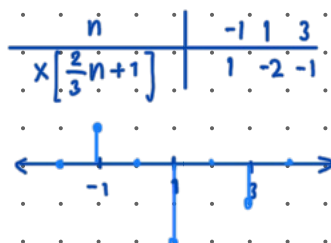
4.1 $x[2-n]$

n	4	3	2	1	0	-1
$x[2-n]$	-3	1	2	-2	3	-1



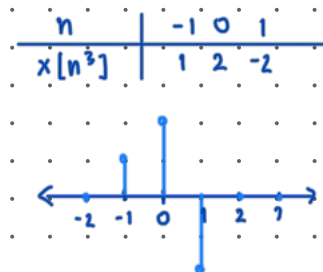
4.3 $x\left[\frac{2}{3}n+1\right]$

This signal will be valid only if $n \cdot \frac{2}{3} = 0$, which results to an integer.



4.5 $x[n^3]$

Only -1, 0, 1 is in the range $[-2, 3]$ when cubed.



// Method provided by Claude

9. Evaluate the following integrals

9.1 $\int_{-\infty}^{\infty} \left(\frac{2}{3}t - \frac{3}{2}\right) \delta(t-1) dt = \frac{2}{3}(1) - \frac{3}{2} = -\frac{5}{6} *$

9.2 $\int_{-3}^{-2} \left[e^{-(t+1)} + \sin\left(\frac{2\pi t}{3}\right)\right] \delta\left(t - \frac{3}{2}\right) dt = 0 *$

Notice that $\frac{3}{2} \notin [-3, -2]$